

Salvador de Sousa Carvalho

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ABOUT

I am a 3rd-year Electrical and Computer Engineering student at Instituto Superior Técnico with a passion for solving problems using different skills. My academic journey is driven by a deep interest in Robotics, Home Automation, and IoT, where I actively explore the intersection of hardware and software to build smarter systems. Outside the classroom, I am a dedicated enthusiast of Lego Technic and Formula 1, hobbies that from young develop my interest for engineering and knowing how things works and are built.

With a C1 Advanced certification in English from Cambridge and currently expanding my linguistic skills by learning German, I am prepared to collaborate in international and multicultural environments. I am always looking for new challenges that allow me to apply my technical background to real-world projects, whether they involve developing autonomous systems or optimizing automated environments.

EXPERIENCE

Formula Student Lisboa, Lisbon, Portugal Jan 2025 – Aug 2025
Trainee, Electronics and Software Department

- Designed and developed electronic schematics and PCB layouts for the vehicle's braking system sensors using **Altium Designer**.
- Programmed microcontrollers in **C** and **C++** to process sensor data and implement real-time control logic for the braking system.
- Performed high-precision **soldering** and hardware assembly, ensuring the reliability of electronic components in high-performance environments.
- Collaborated in team communication and technical presentations, effectively reporting progress and system design specifications.

EDUCATION

Instituto Superior Técnico, Lisbon, Portugal Sep 2023 – Present
Bachelor of Science in Electrical and Computer Engineering

Colégio de São Miguel, Fátima, Portugal Sep 2020 – Jun 2023
Secondary Education, Ciências e Tecnologias **GPA: 17.2/20**
National Mathematics Exam: 18.1/20

Colégio do Sagrado Coração de Maria, Fátima, Portugal Sep 2017 – Jul 2020
3º Ciclo

Colégio de São Miguel, Fátima, Portugal Sep 2015 – Jun 2017
2º Ciclo

EB1 Reguengo do Fetal, Batalha, Portugal Sep 2010 – Jul 2014
1º Ciclo

PROJECTS

Algorithms and Data Structures Projects ([GitHub](#)) 2025

- Developed a series of high-performance applications in **C** focused on algorithmic efficiency and memory management. Implemented complex data structures including **Min-Heaps**, **Linked Lists**, and **Hash Tables**, and applied graph algorithms such as **Dijkstra** for shortest-path problems.

Focused on optimizing time and space complexity (O -notation) to handle large datasets, following the rigorous standards of Instituto Superior Técnico.

CS2 & Faceit Discord Bot ([GitHub](#))

2025

- Developed a custom Discord bot using **discord.py** and **aiohttp** to integrate Faceit API data and live CS2 server status via the A2S protocol. Implemented real-time Elo tracking, match statistics retrieval, and an interactive server browser menu, alongside a voice-system module utilizing **FFmpeg** and **PyNaCl** for custom audio playback. Managed asynchronous backend tasks and environment configurations for seamless community interaction.

5-DOF Robotic Arm ([GitHub](#), [Video](#), [Report](#))

2023

- Developed a 5-DOF robotic arm control system using Arduino and C++, featuring a Serial interface for real-time joint manipulation and user interaction. Implemented input validation and safety constraints for servo positioning via PWM to ensure smooth kinematic movement and hardware longevity. Documented the complete electronic circuit design and firmware architecture in a detailed technical report.

HONORS & AWARDS

Merit Board, Colégio do Sagrado Coração de Maria	2017/18, 2018/19, 2019/20
Honor Board, Colégio do Sagrado Coração de Maria	2017/18, 2018/19, 2019/20
Merit Board, EB1 Reguengo do Fetal	2013/14

CERTIFICATES

Cambridge Advanced English, Cambridge English	Aug 2023
Cambridge First Certificate in English, Cambridge English	Jul 2022

SKILLS AND INTERESTS

Hard skills:

- Programming: C/C++, MATLAB, Simulink, Fusion360
- Tools & Frameworks: Git, Docker, Linux, LTSPICE

Soft skills: Analytical & Creative Thinking, Team Collaboration, Adaptability & Communication

Language skills: Portuguese (native), English (C1), German (A1)

Interests: Lego and Formula1